

Target Tracking for Sensor Networks: A Survey

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Target-tracking algorithms typically organize the network into a logical structure (such as tree, cluster, or faces) to enable data fusion and reduce communication costs. These algorithms often predict the target's future position. Besides using position forecasts for decision making, we can also use such information to save energy, by activating only the set of sensors nearby the target's trajectory. In this work, we survey of the state-of-the-art of target-tracking techniques in sensor networks. We identify three different formulations for the target-tracking problem, and classify the target-tracking algorithms based on common characteristics. Furthermore, for the sake of a better understanding of the target-tracking process, we proposed to organize this process in six components: target detection, node cooperation, current-position estimation, future-position estimation, energy management, and target recovery. Each component has different solutions that affect the target-tracking performance.

Categories and Subject Descriptors: ... [...]: ...

General Terms: Sensor Networks, Distributed Algorithms, Data Fusion

Additional Key Words and Phrases: Target Tracking, Particle Filter; Kalman Filter

1. INTRODUCTION

A Wireless Sensor Network (WSN) [Akyildiz et al. 2002] is an ad-hoc network composed of resource-constrained devices [Nakamura et al. 2014], called sensor nodes. These sensors are able to perceive the environment, collect, process, and exchange data. A WSN may be designed with different objectives, such as event monitoring, data gathering, environment actuation, and target tracking [Nakamura et al. 2007].

Target tracking is an important and non-trivial application of WSNs [Li and Zhou 2010]. The objective is to gather sensor data about one or multiple targets (mobile entities), then estimate their current and future positions in the sensor field [Nakamura et al. 2007]. Target tracking has different areas of application, such as surveillance, habitat monitoring, traffic monitoring, and intruder tracking [Semertzidis et al. 2010; Komagal et al. 2012; Kozma et al. 2012]. Target-tracking algorithms aim at maintaining accuracy and reporting the results quickly, while reducing communication costs and energy consumption [Souza et al. 2011; Hajjaghajani et al. 2012; Hsu et al. 2012; Dongmei and Jinkuan 2013].

In general, the target-tracking problem consists of a WSN, whose nodes are strategically or randomly deployed across the sensor field [Souza et al. 2011; Shi and Tan 2009; Boutaba et al. 2014]. A node detects the presence of targets by

sampling the sensed signals (e.g., light, sound, image, or video) [Gustafsson and Gunnarsson 2007]. Then, these readings are used to estimate the trajectories of one or more targets, and notify the sink node [Zhu et al. 2011]. The target movements are described by mobility models [Vasanthi et al. 2011] that match the type of the target being tracked [Yen et al. 2010].

Many applications demand predicting the target’s position [Nakamura et al. 2007; Sharma et al. 2011b; Kumar and Sivalingam 2012]. This prediction can be used, for instance, to intercept an animal that is moving towards a risky area, such as roads around a forest reserve. However, internal tasks can also use such information to save energy by activating only a set of sensors located within the region where the target is moving to [Xu et al. 2004a; Bhuiyan et al. 2010]. In this case, if the wrong group of sensor is active, the target will be lost, and a target recovery mechanism should be triggered [Khare and Sivalingam 2011].

In this process, we identify six major challenges related to target tracking applications: (1) Target Detection – sensor nodes detect the target; (2) Node Cooperation – nodes cooperate and fuse data, reducing the amount of messages; (3) Position Computation – the data sensed by nodes are fused to determine the target’s position; (4) Future-Position Estimation – the target’s future position is estimated based on historical data of the target movement; (5) Energy Management – a small set of nodes remains active to detect the target at a particular time to save energy; (6) Target Recovery – if the prediction fails, and the target trace is lost, the active nodes perform a search in the network to find the target.

There are several target tracking strategies for these challenges in WSNs. Despite the general goal of these algorithms is estimating the position of an object of interest, these algorithms can be distinguished in several aspects. This differentiation occurs mainly by a specific demand of the application, which modifies the network structure, the communication cost and the operation of the algorithm as a whole. For instance, some applications request that the target’s position is reported at regular intervals [Hajiaghajani et al. 2012], while other applications demand this notification just after an user query [Liu et al. 2008]. This difference in the notification process leads to the use of different tracking algorithms and techniques. In this work, we aiming at highlighting the similarities and differences among target tracking algorithms. Hence, we survey the state-of-the-art related to target-tracking techniques in WSNs, and classify the algorithms based on different criteria, such as problem formulation, algorithm’s components, evaluation metrics, network structure, number of targets, and type of targets.

The remainder of this work is organized as follows. Section 2 presents the terminology and elements used in target tracking. Section 3 divides the target-tracking process into six components, explaining the different techniques employed by each one. Relevant metrics for analyzing target-tracking algorithms are presented in Section 4. The requirements necessary to reach a satisfactory operation of the target-tracking application are presented in Section 5. Section 6 presents a classification of target-tracking algorithms. Finally, Section 7 presents the final remarks.

2. FUNDAMENTALS

The terms target tracking or object tracking are used to describe different formulations of the tracking problem. These formulations change the operation of the algorithms and the elements involved in the process. This section presents the elements involved in target tracking and the different formulations of this problem.

The terminology used for the elements involved in tracking, as well as the elements themselves, can change depending on the characteristics of the tracking algorithm, namely the problem formulation (Section 2.1), the structure used (Section 3.2), and operating mode of the nodes (Section 3.5). Table I details the terms used by target-tracking state-of-the-art.

Table I. Elements of target-tracking algorithms.

Graphical	Element	Others Terms	Description
	Target	Mobile object, moving object	This is the element that the application should track, it is usually a person, animal or vehicle. Depending on the application, more than one target can be tracked simultaneously.
	Active node	Awake node	An active node is capable of sensing the target and communicating with other active nodes within a certain range.
	Node detecting	-	It is the state of the active node that detects the event (target). In this state, the node collects data about the target.
	Inactive node	Sleep node	This is a sensor node that has turned its radio and sensors off. Nodes remain in this state to save energy and can be activated at specific time intervals.
	Cluster head	Group leader, master node	In cluster-based tracking algorithms, the cluster head is responsible for collecting data of its cluster, then perform data fusion.
	Beacon node	Anchor, locator, or seed	In the guided formulation, the beacon node is used to provide the tracker with a route to be followed. Beacon nodes are marked according to the trajectory of the target, building a path that is followed by the tracker.
	Tracker	Interceptor, mobile user, mobile source, guided vehicle	The tracker (a person or a vehicle) in the guided navigation formulation receives the position of the target. The goal is to reach the target in the shortest possible time.
	Sink node	Base station, querying point, processing center	This is the intermediary node between the network and application user. Generally, it is a special node with higher processing power and storage. It collects and stores data from the network and can perform data fusion.

2.1 Problem Formulation

The main purpose of the target-tracking applications is basically the same: estimate the trajectories of one or more targets. However, the way the problem is formulated determines the way the algorithm operates. The target-tracking problem formulations can be classified as: push-based, poll-based, and guided formulations.

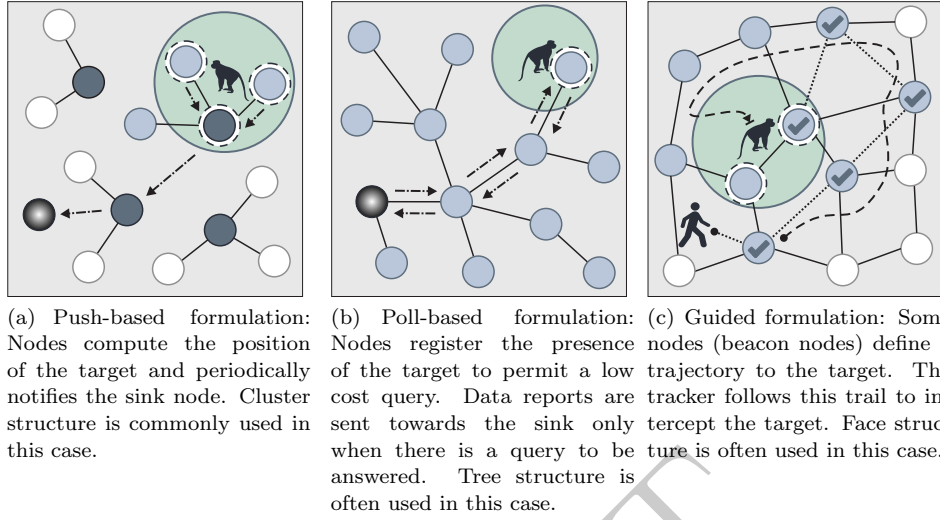


Fig. 1. Tree different types of the target-tracking problem formulation.

The push-based formulation (Figure 1(a)) is the most general tracking technique. The sensor nodes collaborate among themselves to fuse data about the target and generate data reports that are periodically sent towards the sink node. A higher reporting frequency is set when an application needs timely updates about the moving targets. Otherwise, a low reporting frequency can reduce the transmissions, and increase the network lifetime [Xu et al. 2004b; Hajiaghajani et al. 2012].

In the poll-based formulation (Figure 1(b)), data reports are sent to the sink (querying point) only when the sink makes a query. The network acts as a distributed database, in which the nodes register information about the presence of the target to enable a low cost query. In this case, the tracking algorithm aims at an efficient query, by using the least amount of communication to update sensor nodes [Kung and Vlah 2003; Lin et al. 2006; Liu et al. 2008].

In the guided formulation (Figure 1(c)), a tracker is added to the problem. The goal of the tracker is to intercept the target. The tracker can be a person or a guided vehicle that receives the position of the target from the network, so it can move towards the target. The objective is to reach the target in the shortest possible time [Tsai et al. 2007; Bhuiyan et al. 2010; Hsu et al. 2012].

3. COMPONENTS OF THE TARGET-TRACKING ALGORITHMS

The components of target-tracking algorithms (Figure 2) are responsible for the following functionalities: target detection, node cooperation, current-position estimation, future-position estimation, energy management, and target recovery.

3.1 Target Detection

Event detection is an essential element for various sensor network applications. An event is detected when sensor readings match the conditions that describe the event. In target tracking, an event occurs when the target is detected by one or

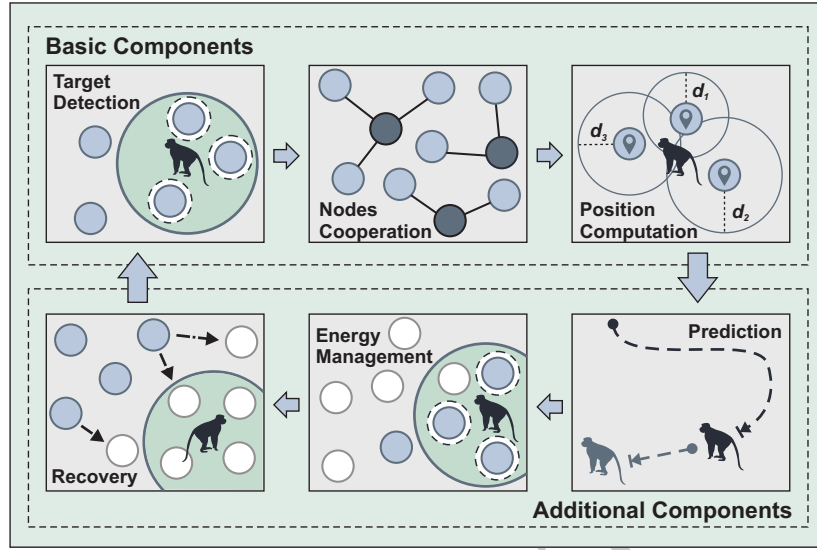


Fig. 2. Components of target-tracking algorithms. Target Detection: sensor nodes detect the target. Nodes Cooperation: nodes cooperate and fuse data, reducing the amount of messages. Position Computation: the data sensed by nodes are fused to determine the position of the target. Future-Position Estimation: the future target position is estimated based on historical data of the target movement. Energy Management: a small set of nodes remains active to detect the target at a particular time to save energy. Target Recovery: if the prediction fails, and the target trace is lost, the active nodes perform a search in the network to find the target.

more nodes [Vairo et al. 2010; Xue et al. 2011]. In practice, the detection process analyzes sensor readings. The type/nature of signals depend on the types of targets being tracked [Gustafsson and Gunnarsson 2007; Jin et al. 2012].

Detection models are used to validate algorithms for WSNs, so algorithms' behaviors can be assessed considering different environmental conditions [Nakamura and Souza 2010].

3.1.1 Sensors and Samples. Some sensor data used for detecting a target include: acoustic, seismic, infrared, magnetic, radio frequency identification (RFID), light, radar, image, and video data.

Acoustic sensors can be used to distinguish objects with different sound signatures [Sleep 2013; An et al. 2014]. Chen et al. [2004] propose to track targets by sensing and analyzing the energy level of acoustic signals to classify the sound. A sensor node accuses detection of the target when acoustic-signal strength exceeds a predetermined threshold. Isbitiren and Akan [2011] propose an alternative underwater target-tracking technique for traditional sonar arrays that use underwater acoustic sensors. The technique is based on echoes from the targets in response to acoustic pulses sent by the sensors.

Vibration sensors can distinguish objects with different weights and speeds. Jin et al. [2012] make use of seismic and passive infrared sensors for target detection and classification of humans, light vehicles, and animals. An algorithm captures the essential signatures of the samples in the time-frequency domain by using a wavelet

feature extraction method. This method demands low processing and memory resources, making it feasible in practical applications. However, PIR sensors deployment can be expensive, because of their short detection range. Wahlström et al. [2011] use magnetometers to detect metallic targets (vehicles), achieving a good tracking accuracy, but its evaluations consider just a single target.

Oka and Lampe [2010] integrate RFID with WSNs for target-tracking purposes. Each target carries an active RFID tag that enables individual target identification. Walchli et al. [2007] show a person tracking algorithm that uses luminosity sensors. In this case, the target must be equipped with a light source, such as a flashlight. The intensity of the light measurements are used for decisions making, leader election, and localization.

Chiani et al. [2009] proposed a radar-sensor network for detecting targets based on impulse radio ultra-wideband. The network is composed of one transmitting node and several receiving nodes, and the sensor field is divided into pixels. Each receiving node computes for each pixel a soft image related to reference pulse, then a sink node combines the correlation maps provided by all receiving nodes. The pixel with the maximum combined metric determines the target location. This approach suffers from high complexity to compute the soft metrics for every pixel. Kozma et al. [2012] describe the integration of miniature low-power radars with acoustic and seismic sensors for detecting, identifying, and tracking vehicles.

Lee and Choi [2011] proposed an algorithm for tracking deformable targets in video images. The detecting procedure extracts moving targets by motion segmentation between frames and tracks moving region detected using optical-flow based detector. Komagal et al. [2012] proposed techniques for target detection in video images. These techniques allow the system to ignore the background motion and identify foreground elements (targets). The advantage of these methods is that the spatial relations of the different features are preserved at the image level. Their disadvantage is that more dimensions requires more training data to obtain a good detector.

3.1.2 Event-Detection Models. Event-based approaches for WSNs must consider event-detection model. The detection efficiency depends on several factors such as environmental conditions, sensor reliability, and event characteristics [Nakamura and Souza 2010]. Thus, event-detection models seek to include the interferences of these factors in their features [Pujolle et al. 2009; Aghaeipoor et al. 2014].

The simplest model is the binary detection [Wang et al. 2010]. In this model, for a given event ϵ , every sensor s , whose distance d is smaller than a detection radius R , detects the event. Then event detection probability is defined as

$$P(s, \epsilon) = \begin{cases} 1, & \text{if } d \leq R \\ 0, & \text{otherwise} \end{cases} . \quad (1)$$

The binary detection model is generally used to evaluate algorithms under ideal conditions. Wang et al. [2010] and He et al. [2010] introduce target-tracking algorithms using binary detection in sensor networks.

The probabilistic detection models include parameters that represent the sensor technology and environmental factors (α and β). The event detection-probability is inversely proportional to a linear [Lu and Suda 2003] or an exponential [Dhillon

et al. 2002] function of the distance separating the sensor and the event:

$$P(s, \epsilon) = (1 + \alpha d)^{-\beta}, \quad (2)$$

and

$$P(s, \epsilon) = \epsilon^{-\alpha d}. \quad (3)$$

Huang et al. [2011] introduced a probabilistic target coverage to detect multiple targets by multiple sensors cooperatively and simultaneously, in which each target is given a detection probability threshold.

Zhang et al. [2006] extended the probability models by adding the influence of the time event duration t in target-tracking applications. The idea is to increase the detection probability when the event remains for a long time at the same position. Thus, detection probability is defined as

$$P(s, \epsilon, t) = \begin{cases} P(s, \epsilon), & \text{if } 0 < t \leq T \\ 1 - (1 - P(s, \epsilon))^{\lfloor t/T \rfloor}, & \text{if } t > T \end{cases}, \quad (4)$$

in which T is the time period during that the sensing algorithm is executed.

A hybrid detection model that merges the binary and probabilistic exponential detection model is introduced by Zou and Chakrabarty [2004]. This model is based on a threshold r , and considers three situations:

$$P(s, \epsilon) = \begin{cases} 1, & \text{if } d < r \\ 0, & \text{if } d > R \\ \alpha e^{-\beta d}, & \text{if } r \leq d \leq R \end{cases}. \quad (5)$$

Nakamura and Souza [2010] proposed a flexible event detection that enables modeling of different scenarios of detection adopted in WSNs, such as fixed radius and exponential models. Thus, event detect probability is defined as

$$P(s, \epsilon) = \alpha \gamma^{(\beta d)^\theta}, \quad (6)$$

in which $0 < \alpha \leq 1$ is a saturation or accuracy parameter, $\gamma > 1$ and $\beta > 0$ are, respectively, the vertical locale and horizontal locale parameters, and $\theta > 0$ is the slope parameter. The experimental validation of this model is performed in a target-tracking task.

3.2 Node Cooperation

Node cooperation defines how communication between nodes occur once an event is detected [Liu et al. 2008; Wang et al. 2010]. This process reduces the number of messages in order to save energy, since it facilitates data aggregation. The majority of target-tracking approaches aim at improving this component [Nakamura et al. 2007; Bhuiyan et al. 2010; Deldar and Yaghmaee 2011].

Naive target-tracking methodologies employ a centralized approach, in which the sensor nodes that detect the target send their data towards the sink node. In this case, the sink node is responsible for fusing the data received, and estimating the target position. As the number of sensors increases, more messages are sent towards the sink. Furthermore, several messages are lost, because of the communication overload [Souza et al. 2009; Singh et al. 2011].

For these reasons, the sensor nodes that detect the target cooperate to send only a single data report to the sink. To make collaborative data processing easier in tracking applications, the logical structure of the sensor network is usually organized as a tree [Lin et al. 2006; Yen et al. 2010], cluster [Xu and Lee 2007; Mansouri et al. 2011], or faces [Huang et al. 2005; Tsai et al. 2007; Ji et al. 2009; Bhuiyan et al. 2010; Hsu et al. 2012; Wang et al. 2014].

3.2.1 Tree-Based Cooperation. In the tree-based cooperation, sensor nodes are organized as a tree or graph, in which the vertices are the sensor nodes and the edges are the communication links (Figure 1(b)).

The tree-based structure can be used in different ways. When the sink node is the root, it finds the shortest path to every node, which costs one network flooding. The network can be reconfigured whenever the topology changes, e.g., sensors that run out of battery or changed position [Tran and Yang 2006; Souza et al. 2011].

Another possibility is to use a dynamic moving tree to track targets. When the target is first detected, the initial tree is constructed by the sensor nodes that detected the target. The node closer to the target is usually the root node. Sensor nodes are added (tree expansion) to the tree as the target approaches them. Sensor nodes can also be removed (tree pruning) from the tree as the target moves away from them. The root node also needs to be changed with the trajectory of the target to optimize the communication overhead. To this end, a simple selection criterion is to choose the node closer to the centroid of the nodes in the tree. Thus, the tree should have a lower communication cost during data collection. In order to send event data to the sink, the leaf nodes send their data reports to their parents. The intermediate nodes fuse their data with data received from their children to generate new single data reports that are relayed to their parents. The root node receives all data reports and generates the final data report that is sent to the sink node. The tree-based tracking algorithms reduce network traffic, redundancy in data transmission, and energy consumption [Zhang and Cao 2004a; 2004b].

Tree-based cooperation can also organize the network as a distributed database, since each sensor node registers when the target enters or leaves its detection range. This structure is called message-pruning tree. The hierarchy is used to connect the sensors by using the sink node (querying point) as the root. In this approach, the tracking application involves two operations: update and query. Updates of a target position are made when the target moves from one sensor to another. A query is always performed whenever it is necessary to find the position a target. The goal is to maintain sensor nodes updated about the target position to allow for efficient querying. A query is routed from the top of the hierarchy, following a single path towards the sensor that reported the wanted target. Without the update of the sensor nodes, the queries would need to be flooded to all leaves, thereby increasing communication cost. The message-pruning tree should permit to update the database and query the target position at minimum cost, but these methods usually need *a priori* mobility models of the tracked targets to achieve efficiency [Kung and Vlah 2003; Lin and Tseng 2004; Lin et al. 2006; Liu et al. 2008; Liu 2010].

3.2.2 Cluster-Based Cooperation. In target-tracking applications, the cluster-based cooperation is used to facilitate collaborative data processing, and to manage the resources of sensor networks to save energy. Clustering is useful when an application requires high scalability of nodes, because it supports data fusion and reduces communication overhead.

Sensor nodes are organized into clusters. Each cluster has a Cluster Head (CH) and, possibly, multiple Cluster Members (CM). A CH is responsible for collecting data from its CMs, performing data fusion and sending a single data report towards the sink (Figure 1(a)). Cluster-based structures are divided into three approaches: static [Olule et al. 2007; Xu and Lee 2007; Deldar and Yaghmaee 2011; Bhatti et al. 2011], dynamic [Ji et al. 2004; Chen et al. 2004; Jin et al. 2006; Yang et al. 2007; Lee et al. 2007; Suganya 2008; Jian et al. 2011], and hybrid (static/dynamic) [Chang et al. 2008; Wang et al. 2010; Hajiaghajani et al. 2012; Feng et al. 2015].

In the static cluster approach, clusters are organized at the time of network deployment, i.e., before target tracking begins. The attributes of each cluster, such as size and its members, are fixed. However the role of being a CH may be rotated periodically among the nodes of the cluster to distribute the energy consumption over all nodes in the network. Selection of CH affects the energy efficiency of the network, thus several strategies considering residual energy and localization of the nodes are studied [Deosarkar et al. 2008; Koucheryavy and Salim 2009]. Some tracking algorithms consider that the clusters are built using hierarchical routing techniques [Chang et al. 2008], such as LEACH [Yektaparast et al. 2012] and PEGASIS [Chen et al. 2011]. Although the static clustering technique is simple and reduces the communication overhead, it presents low tolerance to failures and nodes in distinct clusters cannot share information, causing a boundary problem [Wang et al. 2010]: when the target crosses the boundary of two clusters.

In the dynamic cluster approach, clusters are formed as the target moves through the sensor field. A node is not restricted to a single cluster, it can be a member of different clusters along time. Cluster formation is triggered by the target presence. Nodes that detect the target exchange messages among themselves to decide which one will be the cluster head [Chen et al. 2004]. The dynamic clustering approach minimizes target localization errors, since all data fusion is processed by a single cluster head; unlike the static approach, in which multiple cluster heads may fuse data. In this case, the sensor nodes belong to distinct clusters, decreasing the certainty of the localization, thus the collaboration among them is incomplete and unreliable [Wang et al. 2010]. The requirement of a cluster head election mechanism increases energy consumption, since this mechanism should be performed several times during target tracking [Alaybeyoglu et al. 2009].

The hybrid cluster approach distributes the task of tracking between dynamic and static clusters. The goal is to combine the simplicity and energy savings of static cluster with the flexibility and quality of dynamic cluster techniques. Thus, static clusters are created at the time of network deployment. If an event occurs within the limits of a given cluster, the cluster itself performs the fusion of data generated. On the other hand, if an event covers two or more clusters, these are joined to form a dynamic cluster. Compared to pure dynamic-cluster formation, creating a dynamic cluster from static clusters is cheaper, since this approach reduces the

communication cost and, consequently, the energy consumption [Chang et al. 2008; Wang et al. 2010; Hajiaghajani et al. 2012; Feng et al. 2015].

3.2.3 Face-Based Cooperation. In the face-based cooperation, the network is divided into faces. A face is a group of nodes organized in a ring (Figure 1(c)). A node does not need to get data about all nodes on the network, it only needs to collect these data of the nodes from adjacent faces [Bhuiyan et al. 2010].

The construction of a face structure has two parts: generating planarized graphs; and identifying the faces in the network [Hsu et al. 2012; Wang et al. 2014]. The first part aims at avoiding cases in which the communication of two pairs of nodes is crossing. Thus, for each pair of neighbor nodes is necessary to verify whether there is an intermediate node, in which the communication range between the two nodes can be divided into two shorter communication ranges through the intermediate node. Usually Relative Neighborhood Graph (RNG) [Wang et al. 2010; Hsu et al. 2012] or Gabriel Graph (GG) [Maignan and Gruau 2011; Tsai et al. 2007] algorithms are used for this task. In the second part, each node needs to be aware of the other nodes in the same face, the right-hand rule [Huang et al. 2005] can be used for this. This rule creates a discovery message on a source node that is passed clockwise to all the nodes of the face. As a discovery message flows in a face, the locations of the nodes that the message has traversed are added to it. The locations of all nodes on the face are collected when the discovery message finishes traversing a face. Then this message traverses the same face again to inform the other nodes about the complete discovery results.

Cluster cooperation usually has good performance and it is widely used in WSN. However, the overlap between two clusters and the cluster head selection mechanism need redundant communication that may result in extra cost [Lee et al. 2007]. Face architecture can efficiently solve these problems, since there is no duplicated data problem [Ji et al. 2009]. Furthermore, the face structure is suitable for any kind of node densities, since it solves the hole and obstacle problems [Tsai et al. 2007]. The drawback of face-based methods is the high communication cost required to build the face structure during the configuration phase of the network.

3.3 Current-Position Estimation

The position computation component deals with finding the position of the target by using the location of the nodes that detected the target, and the distance of these nodes to the target. target-tracking algorithms usually consider that nodes know their locations a priori or they are calculated by using a localization system [Boukerche et al. 2007]. The same techniques used to compute the location of the nodes can be used to compute the position of the target. Thus, this component has two steps: distance estimation; and position estimation.

3.3.1 Distance Estimation. The distance estimation is performed according to the sampling signals emitted from the target and collected by the sensor nodes during the target detection [Xu et al. 2013].

The Received Signal Strength Indicator (RSSI) is used to estimate the distance between the sensor and the target, since the output signal captured by the sensor decreases with the distance between itself and the signal source. In other words, the longer the distance between the node and the target, the lower the signal strength

when it arrives at that node.

3.3.2 Position Estimation. When the nodes that detected the target have enough information about their locations and distances to the target, they can compute the position of the target with trilateration, multilateration, or bounding box [Boukerche et al. 2007; Oliveira et al. 2009].

Trilateration computes the position of the target via intersection of three nodes circles. When more than three nodes detected the target, we can use multilateration to compute the position of the target.

The bounding box technique uses squares instead of circles. The intersection of all bounding boxes can be computed without floating point operations by taking the minimum of the high coordinates and the maximum of the low coordinates of all bounding boxes. The final position of the target is the center of the intersection of all bounding boxes.

3.4 Future-Position Estimation

The future-position estimation component computes future positions of the target [Nakamura et al. 2007; Kumar and Sivalingam 2012]. This information can be used to save energy, e.g., activating elements in areas where the target usually stay [Xu et al. 2004b; Bhuiyan et al. 2010; Hong et al. 2010; Deldar and Yaghmaee 2011].

Estimation methods commonly used for tracking targets in sensor networks are Kalman Filter (KF) [Xu et al. 2012; Wang et al. 2013], Alpha-Beta Filter (ABF) [Sharma et al. 2011b], and Particle Filter (PF) [Gustafsson 2010; Jiang and Ravindran 2011; Kalpana and Sangeetha 2013].

3.4.1 Kalman Filter (KF). The Kalman Filter is a popular and efficient recursive method used to fuse low-level redundant data. This method uses a set of inaccurate measurements observed over time and produces estimates that are more accurate than isolated measurements. The KF method retrieves statistically optimal estimates for systems that can be described by a linear model, and the error can be represented as a white noise [Nakamura et al. 2007].

This method applies a linear operator at each discrete-time increment in the current state to generate the new state. The filter considers measurement noise and, optionally, information about the controls on the system. Then, another linear operator, also subject to noise, generates the observed outputs from the true state.

The KF predicts the state $\mathbf{x}$ of a time k by the state-space model

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k + \mathbf{w}_k \quad (7)$$

with measurements $\mathbf{y}$ represented by

$$\mathbf{y}_k = \mathbf{C}\mathbf{x}_k + \mathbf{v}_k, \quad (8)$$

in which $\mathbf{A}$ is the state transition matrix, $\mathbf{B}$ is the input control matrix that is applied to control vector $\mathbf{u}$, $\mathbf{C}$ is the measurement matrix, $\mathbf{w}$ represents the process noise with covariance matrix $\mathbf{Q}$, and $\mathbf{v}$ represents the measurement noise with covariance $\mathbf{R}$. These noise sources are represented by random zero-mean Gaussian variables.

Knowing the system parameters and the measurement $\mathbf{y}$, $\hat{\mathbf{x}}$ is estimated by

$$\hat{\mathbf{x}}_{k+1} = (\mathbf{A}\hat{\mathbf{x}}_k + \mathbf{B}\mathbf{u}_k) + \mathbf{K}_k(\mathbf{y}_k - \mathbf{C}\hat{\mathbf{x}}_k), \quad (9)$$

in which the Kalman gain $\mathbf{K}$ is determined by

$$\mathbf{K}_k = \mathbf{P}_k \mathbf{C}^T (\mathbf{C} \mathbf{P}_k \mathbf{C}^T + \mathbf{R})^{-1}, \quad (10)$$

while the prediction covariance matrix $\mathbf{P}$ is determined by

$$\mathbf{P}_{k+1} = \mathbf{A}(\mathbf{I} - \mathbf{K}_k \mathbf{C}) \mathbf{P}_k \mathbf{A}^T + \mathbf{Q}. \quad (11)$$

The KF is divided into two recursive phases: predict (time-update) and correct (measurement-update) [Xu et al. 2012]. The predict phase obtains the a priori estimates for the next time step (equations 7 and 8). The correct phase incorporates a new measurement into the a priori estimate to improve a posteriori estimate (equations 9, 10, and 11). These phases form a cycle that is maintained while the filter is fed by measurements.

Since many problems cannot be represented by linear models, algorithms have emerged based on the original Kalman Filter formulation to allow these problems to be treated. The major variations of the Kalman Filter for non-linear problems are the Extended Kalman Filter (EKF) [Aydogmus and Talu 2012; Wang et al. 2013] and the Unscented Kalman Filter (UKF) [Gustafsson and Hendeby 2012]. The EKF is the most popular alternative to non-linear problems. This method uses a linearized model of the process using Taylor series, therefore, this is a sub-optimal estimator. The UKF performs estimations on non-linear systems without the need to linearize them, because it uses the principle that a series of discrete sampling points can be used to parameterize the mean and covariance. The quality of UKF estimates are close to standard KF for linear systems. LaViola [2003], Orderud [2005], and Gustafsson and Hendeby [2012] compare the performance of EKF and UKF in target-tracking applications.

Olfati-Saber [2007] and Olfati-Saber and Jalalkamali [2011] presented a distributed version of the KF. This distributed version consists of a network of micro-KFs. Experiments with target-tracking applications show that the results of the distributed version is as good as the centralized version. These distributed estimators have a disadvantage: between each sensing and local updates all sensor nodes communicate among themselves repeatedly to reach a consensus (a global average of the measurements). In addition, if the field dynamics is fast, they may not reach a consensus.

3.4.2 Alpha-Beta Filter (ABF). The Alpha-Beta Filter is a simplified estimator, similar to the KF, that does not require a detailed system model. This filter assumes that a system is properly approximated by a model of two internal states, in which the first state is obtained by integrating the value of the second state over time. The basic principle of the ABF is adjust the values of the coefficients $0 < \alpha < 1$ and $0 < \beta < 1$ (that give name to the filter) to correct the prediction based on measured value [Sharma et al. 2011a]. Due to the low computational cost, the ABF is a natural choice for target-tracking applications in WSNs.

The ABF approximates the maneuvering of the target by using the variables x (position) e v (velocity) as internal states. The measurement values correspond to

observations of location x_k in k -th moment, plus the measurement noise. Assuming that the velocity remains approximately constant between measurements during the time interval ΔT , the ABF estimates the value of x through equation

$$\hat{x}_k = \hat{x}_{k-1} + \Delta T \hat{v}_{k-1}. \quad (12)$$

Assuming that v is constant, and the projected value at the next sample is equal to the current value, we have

$$\hat{v}_k = \hat{v}_{k-1}. \quad (13)$$

Equation 13 can include any additional information that is discovered during every time interval.

The effects of noise are accounted by using a simplified model. The difference between measured values and predicted values can increase. This prediction error r , called residual error, can be computed as

$$\hat{r}_k = x_k - \hat{x}_k. \quad (14)$$

The constants α and β are used together with r to correct the estimated position and estimated velocity values, respectively.

$$\hat{x}_k = \hat{x}_k + \alpha r_k \quad (15)$$

$$\hat{v}_k = \hat{v}_k + \left(\frac{\beta}{\Delta T} \right) r_k. \quad (16)$$

The equations above form a recursive relationship, like the KF, because the process is repeated when a time increment occurs, i.e., when a new measure is obtained. Equations 12 and 13 are used in the prediction phase to estimate the next (future) position and velocity values. Finally, equations 14, 15 and 16 are used in the correction phase [Sharma et al. 2011b].

The values of α and β are generally empirically adjusted based on the target motion. These coefficients replace the optimized coefficients of the KF [Wang et al. 2007].

The target-tracking approaches proposed by Wang et al. [2007], Sharma et al. [2011a], and Sharma et al. [2011b] are based on the Alpha-Beta Filter. In those references, the authors compare the ABF and KF predictions, reaching the conclusion that, despite the simplicity, ABF can achieve results that are as good as KF, but their analyses are limited, since they consider that the target has a constant speed.

3.4.3 Particle Filter (PF). The PF is a recursive version of Sequential Monte Carlo (SMC) method [Arulampalam et al. 2002]. PFs represents an alternative to the KF for applications with non-Gaussian noise, especially when there is high computational power available and the sampling rate is slow [Nakamura et al. 2007].

Unlike the linear/Gaussian problems, the calculations of the posterior distribution of non-linear/non-Gaussian problems are extremely complex. To overcome this difficulty, the PF adopts an approach called sampling importance. The goal is to estimate the posterior probability density, representing it as a set of particles.

The Particle Filter aims to build the posterior probability density function (PDF) based on its particles, which are a large number of random samples. Alternating between sampling and resampling methods, these particles are propagated over time. At each discrete time step, resampling increases the relevance of regions with high PDF by discard some particles. The particle quality is indicated by a weight associated with it, then the weighted sum of all particles is the estimate [Jiang and Ravindran 2011].

The resampling step is the solution adopted to avoid the degeneration problem, where particles have negligible weights after several iterations. The particles of greater weight are selected and serve as the basis for the creation of a new particles set. Furthermore, the minor particles disappear and do not originate descendants [Jafaryahya et al. 2010].

The PF is divided into two phases: prediction and correction. The prediction phase modifies each particle according to the settled model, this method also adds random noise to the particles in order to consider the effects of noise. The correction phase reevaluates the weights of the particles based on the new measures of the current step, eliminating the particles with small weights.

The Particle Filter is also widely used in target tracking in sensor networks. Arulampalam et al. [2002] show that in most real-world scenarios, the assumptions of the Kalman Filter (scenarios with linear model and white noise) do not hold. In this case, other approximation algorithms, such as the Particle Filter, must be used. However, when this assumptions hold, no other algorithm can outperform it, because KF retrieves the optimal statistical solution in that case.

Aydogmus and Talu [2012], Weng et al. [2010], and Arulampalam et al. [2002] compare the performance of PF and EKF, and show that PF has better results in several real scenarios. The reason is that the EKF always linearize the models to approximate the density to be Gaussian. However, this density can not be well described when the true density is non-Gaussian. The Particle Filter uses a finite number of samples to approximates the density, directly. Thus, the PF produces more accurate estimates in comparison to the EKF.

Djuric et al. [2008], Ahmed et al. [2010], and Hui-Ying et al. [2010] propose target-tracking algorithms for WSNs that use PF. Hou et al. [2010] combines PF with EKF for tracking multiple targets, improving the performance of these isolated filters. Jiang and Ravindran [2011] and Sheng et al. [2005] proposed distributed target tracking approaches of the Particle Filter to minimize the communication cost of the network by distribute the computational workload onto several nodes, however they do not consider the channel imperfections as part of tracking problems.

3.5 Energy Management

The energy management component is responsible for turning the radio on and off to save energy [Zahedi et al. 2010; Deldar and Yaghmaee 2011; Hsu et al. 2012; Dongmei and Jinkuan 2013]. A node is active when its radio is on, and inactive otherwise. Energy saving is the most important design factor for sensor networks, since limited batteries are used to power the sensor nodes [Yan and Wang 2011].

A sensor node radio has four different operating states: transmit, receive, idle or listening (the radio is on, but nothing is transmitted or received), and sleep (the radio is turned off) [Schurgers 2008]. When the radio is on, the power consumption

does not vary significantly. So the only way to reduce the power consumption is to turn the radio off [Miller and Vaidya 2004].

In target-tracking applications, most of the nodes spend a large amount of time in a state which they do not participate actively in communications. In this case, the radio of the node can be turned off to save energy. Cooperative strategies use prediction information to choose a set of nodes to perform the tracking task, and the remaining nodes can be turned off [Yan and Wang 2011]. Based on the prediction result, a sensor uses the awakening mechanism to activate an adjacent sensor node before the target leaves its own detection field and enters an adjacent field [Jin et al. 2006].

Two strategies are used to manage radio operation: duty cycle [Yan and Wang 2011; Zahedi et al. 2010; Nguyen-Trung et al. 2014]; and paging channel [Xu et al. 2004a; Bhuiyan et al. 2010]. Duty cycle defines time slots in which nodes are active during their lifetime. Thus, sensor nodes change between active and sleep state based on network activity. A long duty cycle allows the sensors to have more time to transmit data, and a low duty cycle helps sensors conserve battery power [Hsu and Wu 2008]. The paging channel allows a sensor node to awaken other nodes. It is a low-energy channel for minimizing the energy consumed during idle state. To this end, nodes are equipped with dual radio: a main radio responsible for data transmissions; and a paging radio to awaken nodes that have turned their main radios off. These radios operate in different frequencies to avoid interference [Schurgers et al. 2002].

Hsu and Wu [2008] proposed a duty cycle adjustment to save energy of nodes with low data traffic and decrease transmission latency of sensors with heavy data traffic in which each node has different idle and sleep schedules with different duty cycles. Yan and Wang [2011] used duty cycle and dynamic clustering to extend sensor network lifetime of target-tracking applications. Zahedi et al. [2010] analyzed the impact of different duty cycle mechanisms on target-tracking performance. These algorithms require that sensor nodes are time synchronized, and this synchronization is achieved by exchanging messages, which leads to using more energy.

Xu et al. [2004a], Wang et al. [2010], Deldar and Yaghmaee [2011], and Hsu et al. [2012] use paging channel to turn radios on by using awakening messages. These approaches predict the future position of the target, and use it to provide an awakening method to decide which sensor nodes have to be active for future tracking. The majority of the commercial sensor nodes does not include this feature, so it must be included as an extra component.

3.6 Target Recovery

The target recovery component aims at finding a lost target once failures render a target temporarily untraceable. The challenge lies in finding the target accurately and quickly, while using as few messages as possible [Khare and Sivalingam 2011].

The prediction information is used to choose a set of nodes that will stay in active mode and track the target, while the remaining nodes stay in sleep mode to reduce energy consumption and increase network lifetime. On the other hand, there is no guarantee that the appropriate set of sensors will be awake, which may result in losing the target's trace [Gupta et al. 2012; Zeng et al. 2014]. In this scenario, a recovery mechanism is necessary since there is no means to avoid losing the target

[Hsu et al. 2012].

The quality of the prediction information is responsible for target losses. There are several reasons that affect the quality of the prediction. One reason is the localization errors, because localization information is used to compute and predict the target's position [Souza et al. 2009; Campos et al. 2012; Savic et al. 2015]. Another reason is node failures, since we may not have sufficient data to compute the target's position accurately. This same problem can be caused by communication breakdown. Although the network is working in perfect condition, a sudden change in the trajectory of the target can induce high prediction error [Gupta et al. 2012].

A recovery mechanism consists of waking up a set of nodes larger than a conventional set (sensor set chosen to be awake based on prediction information) to find a lost target. A simple solution is to awaken all nodes in the network, though with high cost as a tradeoff. Therefore, recovery mechanisms are designed to awaken the minimum amount of sensors needed to find the target.

Xu et al. [2004b] and Yang and Sikdar [2003] awaken nodes around the predicted trajectory of the lost target. If the target is not found, the recovery mechanism floods an awakening message to activate all sensors. This kind of recovery mechanism consumes more energy, because of the increased number of active nodes. Lalooses et al. [2007] proposed a recovery approach for wildlife tracking based on favorite places. If the target is lost, its presence is checked in places the animals frequently visit, such as places where they look for water, shelter and rest. However a flooding phase is still necessary when the target is located in other places.

Khare and Sivalingam [2011] proposed a recovery mechanism to find lost targets in clustered sensor networks. It is divided in four phases: The first phase is responsible for finding out if the target is lost. If the cluster head does not sense the target in a certain amount of time, it concludes that the target is lost; The second phase avoids cases of false recovery initiation. Two or more clusters may be predicted as the next possible location of the target. Then, these groups query each other to see if any of them is tracking the target, otherwise the next phase is triggered. In the third phase, the current cluster awakens its one hop clusters to find the target. If necessary, these clusters awaken their one hop clusters. While the target is not found, this process is repeated. In the last phase, once the target is found, clusters that are participating in the current tracking task remain awake and the remaining clusters go to sleep state. Gupta et al. [2012] proposed a recovery mechanism similar to the one proposed by Khare and Sivalingam [2011], however it uses Kalman Filter predictions to select the clusters that should be awake.

In Ji et al. [2009], when the target is lost, the last node that detected the target awakens all its adjacent faces. If the target is not found, a message is sent to the sink and the tracking is restarted. Hsu et al. [2012] proposed two recovery mechanisms that can be adopted depending of the application needs. The first is used to find the lost target as soon as possible based on the maximum moving speed to awake the sensor nodes close to the lost location. The second is used to find the target by using the least amount of communication, it sequentially searches the target at all faces of the sensor field, beginning with those most likely to contain the target.

4. METRICS

The choice of metrics for analyzing target-tracking algorithms depends on the tracking goals. The metrics such as precision, delay, overhead, and energy consumption are fundamentals in tracking algorithms.

4.1 Precision

Tracking precision measures the difference between the estimated and the actual target's position [Souza et al. 2009; Souza et al. 2011]. This metric is also applied to the predicted position of the target. In this case, the predicted position is compared to its real position at a particular time to estimate the prediction error [Bhuiyan et al. 2010; Hajiaghajani et al. 2012; Cai et al. 2014].

Thus, the measure error ϵ of a 2D target position can be computed as

$$\epsilon = \sqrt{(x_e - x_a)^2 + (y_e - y_a)^2} \quad (17)$$

in which (x_e, y_e) is the estimated target position, and (x_a, y_a) is the actual target position. Similarly, the prediction error can be computed by using the predicted position instead the estimated position.

The precision of the predicted position is related to the energy consumed by the network during target tracking. The energy management component uses predictions to choose the set of sensors that will be used to track the target. An inaccurate prediction can activate the wrong set of sensors and probably end up losing the target. Thus, to recover the target's trace, we need to activate additional nodes and exchange more messages [Xu et al. 2004b; Bhuiyan et al. 2010; Hsu et al. 2012].

4.2 Delay

Delay is an important metric for real-time tracking applications [Wei et al. 2013]. This metric accounts the time the system takes to estimate/predict the target's position. The time taken to send the data towards the sink can be accounted as well [Souza et al. 2011].

There are several factors that may cause delays in tracking applications. To avoid packet collisions in sensor networks, nodes usually wait a small random time before sending a message. Delay accumulates as the message traverses the networks in a multi-hop fashion, resulting in a significant delay to reach the sink [Sharma et al. 2011b; Souza et al. 2011]. Holes or obstacles in the network, caused by dead nodes or rough terrain, can also cause delay, since they can increase the number of hops required to reach the sink [Olule et al. 2007].

In poll-based target-tracking approaches [Kung and Vlah 2003; Lin et al. 2006; Liu et al. 2008; Sharma et al. 2014], delay is defined as the time required for notifying the target's position to the sink (querying point). In this case, the largest delay occurs in a degenerate tree. Thus, we should keep the tree balanced to reduce delay.

In case of losing the target's trace, the target tracking must be recovered as soon as possible. The delay may be defined as the time that it takes to recover the target trace after the recovery mechanism has started [Hsu et al. 2012].

To create or update a dynamic cluster, the nodes usually need to wait for a random time for getting information from other nodes to elect a new cluster head

[Jin et al. 2006; Lee et al. 2007; Hamouda and Phillips 2011; Park et al. 2010; Hajiaghajani et al. 2012]. Each node that detects the target approaching to the boundary of distinct clusters reports this event to their cluster head. Then, the cluster heads of these clusters exchange information with each other to compute the final estimation. This process causes a large time delay [Wang et al. 2010].

The delay δ to notify an event e to the sink node can be expressed as

$$\delta_e = r_e - d_e, \quad (18)$$

in which r is the instant when the sink node is notified about event e , and d is the instant that the event e is detected by a sensor node.

4.3 Communication Cost

Communication cost accounts the amount of messages necessary to setup the network and track the target. The goal of tracking algorithms is to maintain tracking quality, while reducing communication costs.

There are several factors that can result in communication overhead, such as network structure, data precision, data reporting frequency, and number of targets. To avoid that all sensor nodes that detect the target send their sensed data to the sink, target-tracking algorithms can organize the network structure as a tree [Sharma et al. 2014], cluster [Hajiaghajani et al. 2012] or face [Hsu et al. 2012], to enable data fusion and reduce communication cost.

Higher data precision demands more sensor data, which results in communication overhead. A higher data reporting frequency is set when an application needs timely updates about the target, whereas a low reporting frequency reduces the number of transmissions. In these cases, tracking quality and communication cost need to be evaluated [Xu et al. 2004b].

The number of targets tracked can increase network traffic significantly. In this case, nodes can fuse the collected data to reduce traffic when there is more than one target in the same area [Hamouda and Phillips 2011; Hajiaghajani et al. 2012].

The message complexity of a protocol is defined as the total number of messages sent by all nodes. In a flooding communication scenario, to complete one flooding, the message complexity is $O(n)$, because all n nodes have to send a message. In this way, to complete m flooding processes is needed to send $O(m \times n)$ messages.

4.4 Energy Consumption

Energy consumption is a key factor for any application in sensor network, since sensors have energy restrictions. This metric summarizes the amount of energy used by nodes in the network to perform the tracking task [Deldar and Yaghmaee 2011; Hamouda and Phillips 2011].

A typical node is equipped with battery, a radio, multiple sensors, and processing unit. So the energy consumed by a sensor node consists of receiving, transmitting, listening messages on the radio channel, sampling data, and computing. Once energy is over, a sensor node becomes unusable if we cannot replace batteries. Furthermore, to increase network lifetime, we need to save power while performing the tracking task [Yan and Wang 2011].

The communication module is responsible for much of the energy consumption. Therefore, tracking algorithms use data fusion to reduce the number of transmitted

messages [Nakamura et al. 2007; Sleep 2013]. These algorithms also use an awakening mechanism to activate only the set of sensor nodes needed to track the target, while the remainder keep sleeping to save energy [Zahedi et al. 2010; Wang et al. 2014; Zeng et al. 2014]. Moreover, prediction errors may lead to trace lost. To find the target again, we need to awaken several sensors and transmit extra messages, thereby increasing energy consumption. Thus, the accuracy of the prediction is important to extend network lifetime [Khare and Sivalingam 2011; Hsu et al. 2012; Gupta et al. 2012].

The power consumption of data acquisition and processing cannot be neglected. The intensive use of sensors to monitor complex phenomena in practical scenarios can consume as much energy as the communication module or even more. As a consequence, energy management schemes need to be complemented with techniques at the sensor level. Intuitively, these techniques operate to reduce the number of data samples [Alippi et al. 2009; Razzaque and Dobson 2014].

The energy consumed at the instant t , is given by

$$e(t) = \sum_{i=1}^n e_i(0) - e_i(t), \quad (19)$$

in which $e_i(0)$ is the residual energy in the sensor i at the instant 0, and $e_i(t)$ is the residual energy in the sensor i at the instant t .

5. TARGET-TRACKING REQUIREMENTS

To reach a satisfactory operation of the target-tracking application, some secondary objectives, or requirements, need to be fulfilled. Among other requirements, the network should: establish a localization system; perform data routing; control network density; synchronize the clock of sensor nodes; and provide a secure communication service. Figure 3 illustrates these requirements.

Several solutions that focus exclusively on target tracking assume that these requirements are ideally satisfied. However, it is important to evaluate the integration of these requirements with target tracking, because they can cause errors that affect the final result of the application [Souza et al. 2009; Campos et al. 2012].

5.1 Localization System

A localization system is required to provide sensor nodes with position information. Usually, only a fraction of sensor nodes knows their location a priori (using GPS or manual positioning). These nodes are called beacon nodes. A localization algorithm shares beacon information to estimate the location of other nodes. However, this estimate is not perfect [Boukerche et al. 2007].

Two important localization systems are RPE (Recursive Position Estimation) [Albowicz et al. 2001] and DPE (Directed Position Estimation) [Oliveira et al. 2009]. The RPE is a pioneer iterative solution. In this algorithm, each beacon node broadcasts its position information to assist its neighbor nodes to compute their positions. The estimated positions are also used as references, thus the number of references increases iteratively. A problem with this approach is that it cannot estimate the position of several nodes when the network density is low.

The DPE is a solution that evolved from the RPE by reducing errors and the

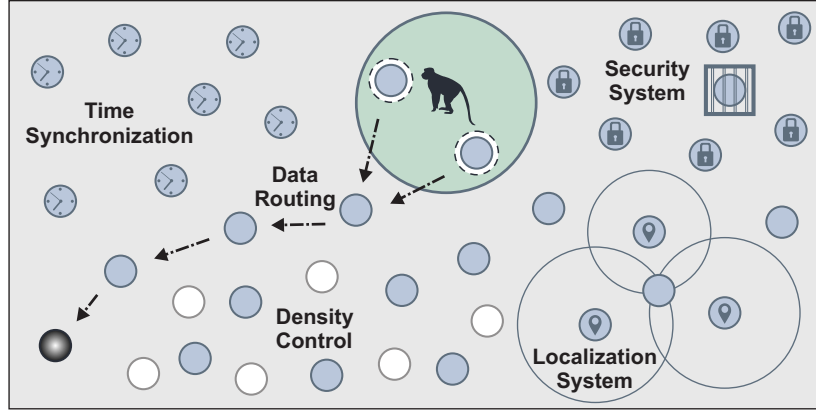


Fig. 3. Requirements for a satisfactory operation of target-tracking applications in WSNs. Localization System: provides the location of nodes; Data Routing: defines appropriate multi-hop paths to send data reports towards the sink node; Density Control: maintains a minimum number of sensor nodes active; Time Synchronization: provides a common reference clock for sensor nodes; Security System: verifies the content of messages and isolates malicious nodes to provide integrity and confidentiality.

overall cost. The main goal of this algorithm is to use a beacon structure to start the recursion at a single point and induce it to follow a specific direction. Thus, only two references are necessary to estimate the position of a sensor node. This approach reduces its accuracy if the beacon structure is far away from the network center.

Target-tracking applications depend on node localization, because sensor nodes' position is used to compute the target's position. However, localization errors have a negative impact in tracking performance [Savic et al. 2015]. Souza et al. [2009] and Campos et al. [2012] evaluated the impact of the actual localization errors in target tracking with Kalman and Particle filters.

5.2 Data Routing

In sensor networks, routing algorithms define multi-hop paths that connect sensor nodes to the sink node, using the smallest possible number of transmissions. Thus, these paths are used to send data reports to the sink node [Al-Karaki and Kamal 2004]. In target-tracking applications, a routing algorithm is an essential player, since it selects appropriate paths to transmit the location of the target to the sink node [Sharma et al. 2011b; Hsu et al. 2012].

Based on network structure, routing algorithms are classified into three categories: flat [Intanagonwiwat et al. 2003], hierarchical [Yektafarast et al. 2012], and geographical [Zhou et al. 2010]. In flat routing, all nodes have the same information about the state of the network and equal roles in gathering information. In a hierarchical, the most powerful nodes (or with more energy) can be used to process and send information, while the simplest nodes can only be used to collect data (sensing). Data fusion can be used to reduce energy consumption by reducing the number of messages sent towards the sink. In geographical routing, the localization of the nodes are exploited to route data through the network. In this case, messages

are often routed in a greedy manner, i.e., a node always forwards a message to the neighbor closest to the sink.

5.3 Density Control

Density control consists in turning off as many nodes as possible, while maintaining the network functionality. The goal of density control algorithms is to improve network lifetime, by activating just a minimum number of nodes needed. If all sensor nodes of a high-density sensor network operate in the active mode simultaneously, the data collected would be highly correlated and redundant, consuming unnecessary energy [Jia et al. 2012].

Shang and Shi [2005] proposed three density control algorithms that consider the trade-off between energy usage and coverage. These three algorithms maximize coverage while avoiding overlap in sensing areas based on localization, distance, and adjustable sensing range. These algorithms request that each node knows its location or the distance to its neighbors, but they do not assess localization errors or distance estimation errors. Kapnadak and Coyle [2011] showed an optimal node density for detecting a target in a sensor field, and Campos et al. [2012] evaluated how different combinations of density control algorithms and localization systems affect target-tracking applications.

5.4 Time Synchronization

Time synchronization aims to provide a common timescale for local clocks of nodes in the sensor network [Lasassmeh and Conrad 2010]. The clock of a sensor node consists of an oscillator and a counter. Based on the oscillator angular frequency, the counter increases its value to represent the local clock. In ideal situations, angular frequency is constant, but due to physical variations (e.g. temperature, vibration, and pressure) this frequency changes and computer clocks drift [Yik-Chung et al. 2011].

In WSNs, we need time synchronization for coordinating nodes to achieve a complex sensing task. Data fusion [Nakamura et al. 2007] is an example of such coordination in which data collected at different nodes are aggregated into a meaningful result. In target-tracking applications, sensor nodes exchange their location and time to identify and estimate the target's position and velocity [Merhi et al. 2009; Dixiao et al. 2010; Bhuiyan et al. 2010]. If sensor nodes are not synchronized, tracking performance is affected. However, there are a few approaches that do not require node synchronization to track a target [Wang et al. 2010; Lee et al. 2011]. Time synchronization can also be used to save energy in tracking applications [Hsu and Wu 2008; Zahedi et al. 2010]. Sensors may sleep at appropriate times and awaken when necessary.

Zhou et al. [2012] improved accuracy and save energy in multiple target tracking by synchronizing the sensors involved in the tracking task. Pashazadeh and Sharifi [2008] evaluated the impact of synchronization errors in tracking applications to aid the application in adjusting the resynchronization time more accurately. Lee et al. [2011] proposed a target-tracking approach that operates independently of the time marked by the sensor nodes. Although these works consider the synchronization problem in their approaches, they do not take into account more realistic and challenging aspects of sensor networks, such as communication delay, sensing errors,

and localization errors.

5.5 Security System

The goal of security systems is to provide confidentiality, integrity, and authenticity for all messages. In WSNs, every sensor node should be able to verify the integrity of all messages it receives, as well as the identity of the sender. Intruders should not be able to read messages' contents [Oracevic et al. 2014]. However, security solutions are expensive for WSNs [Du and Chen 2008].

Security is fundamental for target-tracking applications in military and security monitoring areas. In target tracking, node localization and synchronization are vulnerable to several attacks. For example, before the localization process begins, an attacker may compromise a beacon node to exchange malicious location references [Iqbal and Murshed 2010]. During time synchronization, an attacker may modify timing messages to harm or interrupt the synchronization process [Liu et al. 2012].

To prevent attacks on localization systems in tracking applications, Chang et al. [2007] used the relaxation labeling algorithm to detect malicious nodes, and remove their data from the tracking computations. The authors show that by removing the malicious data, the tracking accuracy is improved, however the authors use only one-dimensional sensor field in low scale, making it hard to know about the scalability of this approach.

Mansouri and Khoukhi [2011] proposed an approach to estimate the position of the target based on quantized proximity sensors by selecting the appropriate group of candidate sensors that participate in data collection. They assume that the communication between active sensor nodes is achieved in a single hop manner, which is not a real scenario for distributed systems of wireless sensor networks.

6. CLASSIFYING TARGET-TRACKING ALGORITHMS

Target-tracking algorithms can be classified by using different criteria, such as network structure, problem formulation, type of the target, and number of targets. Table II shows the algorithms investigated in this work and summarizes our classification proposal. Furthermore, it shows the additional components that are employed by each algorithm.

In the following, we explore the classification of the tracking algorithms in more detail.

6.1 Classification Based on Network Structure

A key criterion for classifying target-tracking algorithms is the structure.

6.1.1 Tree-Based Structure. The Optimized Communication and Organization (OCO) [Tran and Yang 2006] uses a simple tree, in which a flooding started by the sink helps the nodes find the shortest path to the sink. When network topology changes, the tree can be reconfigured.

The dynamic convoy tree-based collaboration (DCTC) [Zhang and Cao 2004a] is a dynamic tree-based structure to enhance the collaboration among the sensor nodes surrounding the target. When the target is first detected, the initial tree is constructed. The root is usually the node closest to the target. Sensor nodes are added to the tree as the target approaches them. Conversely, nodes are removed

Table II. Summary of the classification scheme.

Structure		Method	Additional Components			Target		Formulation
			Prediction	Energy Management	Target Recovery	Multiple	Type	
Tree		DAB – [Kung and Vlah 2003]	no	no	no	yes	simple	poll
		DCTC – [Zhang and Cao 2004a]	yes	yes	no	no	simple	push
		DAT – [Lin et al. 2006]	no	no	no	yes	simple	poll
		OCO – [Tran and Yang 2006]	no	yes	yes	yes	simple	push
		DAT+Shortcut – [Liu et al. 2008]	no	no	no	yes	simple	poll
		MOT – [Sharma et al. 2014]	no	no	no	yes	simple	poll
Cluster	Static	PES – [Xu et al. 2004b]	yes	yes	yes	yes	simple	push
		RARE – [Olule et al. 2007]	no	no	no	no	simple	push
		[Sharma et al. 2011a]	yes	yes	no	no	simple	push
		[Sharma et al. 2011b]	yes	yes	no	no	simple	push
		[Kumar and Sivalingam 2012]	yes	yes	no	no	simple	guided
	Dynamic	[Jin et al. 2006]	yes	yes	yes	no	simple	push
		[Lee et al. 2007]	yes	yes	no	no	simple	push
		ADCT – [Yang et al. 2007]	yes	yes	yes	no	simple	push
		DELTA – [Walchli et al. 2007]	no	no	no	no	simple	push
		PRECO – [Hong et al. 2010]	yes	yes	no	no	continuous	push
		DMMT – [Hamouda and Phillips 2011]	yes	yes	yes	yes	simple	push
		CODA – [Chang et al. 2008]	no	no	no	yes	continuous	push
	Hybrid	HCTT – [Wang et al. 2010]	yes	yes	yes	no	simple	push
		TG-COD – [Park et al. 2010]	yes	no	no	no	continuous	push
		HCMTT – [Hajiaghajani et al. 2012]	no	yes	no	yes	simple	push
Face		DOT – [Tsai et al. 2007]	no	yes	no	no	simple	guided
		FOTP – [Ji et al. 2009]	yes	yes	yes	no	simple	push
		PET – [Bhuiyan et al. 2010]	yes	yes	yes	no	simple	guided
		POOT – [Hsu et al. 2012]	yes	yes	yes	no	simple	guided

from the tree as the target moves away from them.

In DAB [Kung and Vlah 2003], DAT [Lin et al. 2006], and DAT+Shortcut [Liu et al. 2008], the sensor network is considered a distributed database. They use a hierarchical approach to record information about the presence of targets. A message-pruning tree connects all the sensors aiming at reducing redundant transmissions. Tracking consists of two operations: update and query. The detection event follows the tree to update the database when the target crosses from one detection area to another. And the query follows the tree, seeking the sensor closest to the target, when a query for one target is issued at the root. Thus, information is sent towards the sink when requested. It is important for updating the database and querying targets by construct the message-pruning tree of minimum cost. These methods greatly reduce the communication cost through hierarchical organization, however they need to know, *a priori*, the mobility patterns of the targets.

In the DAB, the message-pruning hierarchy tree is based on expected properties of the target movement patterns such as the frequency of target movements over a region. In a DAB tree, an edge consists of multiple communication hops between sensor nodes, therefore it does not represent the network physical structure. Besides, the DAB algorithm does not take into account the query cost, hence it does not minimize the query cost. The message-pruning tree is constructed by DAT by taking the physical topology of the network into consideration. In this approach, update and query costs are associated with the tree construction. Liu et al. [2008] proposed to add shortcuts to the message-pruning tree constructed by DAT to reduce the total cost for updating the database and querying targets. The need for target mobility knowledge in DAB and DAT is not required by the MOT algorithm [Sharma et al. 2014]. This method achieves logarithmic complexity for update cost and query cost, but it assumes that links and sensor nodes do not fail, this assumption is too strong for practical wireless sensor networks applications.

6.1.2 Cluster-Based Structure. In the PES [Xu et al. 2004b] algorithm, the network structure was considered out of scope. Thus, PES assumes that each node operates like a cluster head in a cluster.

The RARE [Olule et al. 2007] algorithm considers a static cluster formation, in which the cluster head is fixed, i.e., it is always the same node, to evaluate energy savings within the cluster. Kumar and Sivalingam [2012] used a static cluster-based network architecture. All sensor nodes were divided into groups and each group was statistically assigned to one cluster head. Sharma et al. [2011a] and Sharma et al. [2011b] proposed a mechanism that forms static clusters that are followed by tree formation over cluster heads, having the sink node as root.

Dynamic clusters are formed according to the presence of the target. In ADCT [Yang et al. 2007], the sensor nodes that detect the target send messages to their neighboring nodes containing their distances to the target. A node becomes a cluster head candidate if it does not receive any message from its neighbors containing a distance shorter than its own distance to the target. These candidates flood a message to define who is the leader. Member nodes join and leave a cluster as the target moves, and prediction information is used to select of a new cluster head. The DELTA [Walchli et al. 2007] algorithm maintains clusters dynamically formed

around a target as soon sensors detect it. A measurement-based election algorithm determines the cluster head. Cluster members wait for periodic messages to be notified about the presence of the cluster head. In the DMMT [Hamouda and Phillips 2011] clustering mechanism, one node acts as the main node, and it is responsible for reforming the cluster if it is necessary. The PRECO [Hong et al. 2010] algorithm constructs a dynamic cluster by selecting the nodes close to the boundary of the continuous target. Lee et al. [2007] proposed a dynamic cluster to reduce redundant data transmissions caused by overlapping areas between clusters. Jin et al. [2006] present a dynamic clustering mechanism for target tracking in WSNs in which cluster formation is based on the route of the target.

Hybrid clusters are formed by using the advantage of static and dynamic clusters. The CODA [Chang et al. 2008] algorithm divides the network into static clusters, where the nodes that detect a continuous target transmit a detection message to their cluster head. With this information, the cluster head estimates the boundary sensors. The boundary sensors of different static clusters are united to form a dynamic group that sends information about the boundary of the continuous target towards the sink. The TG-COD [Park et al. 2010] mechanism constructs static clusters at the network deployment time by dividing the network into coarse-grained grid cells. A dynamic cluster is constructed only when a continuous target appears. In this case, a fine-grained grid is constructed in coarse-grained grid cells. In the HCTT [Wang et al. 2010] and HCMTT [Hajiaghajani et al. 2012] algorithms, boundary nodes of static clusters contribute to dynamic cluster formation. The tracking task is alternated between static and dynamic clusters when the target reaches the edges of static clusters.

6.1.3 Face-Based Structure. The DOT [Tsai et al. 2007], FOTP [Ji et al. 2009], PET [Bhuiyan et al. 2010], and POOT [Hsu et al. 2012] target-tracking algorithms use face structures to distinguish different areas in a sensor network and to limit the number of waking faces as to reduce communication overhead as well as energy consumption.

The DOT and FOTP target-tracking algorithms use planar graphs generated by Gabriel graphs to create their face structures. Finally, the PET and POOT algorithms use RNG to create their faces. In this case, nodes exchange their location information and compute the face neighbors. Each node only gets the information about nodes in its face.

The DOT and PET algorithms keep a node activated whenever a target moves into its face, making a trail of movements of the target. The FOTP combines face structure with hexagon algorithm and predictions to minimize the amount of node serving the tracking task. The POOT shows efficient target recovery for face structures to reduce the time and the number of transmissions to complete this task.

6.2 Classification Based on Problem Formulation

The DAB, DAT, DAT+Shortcut, and MOT are used in the poll-based target-tracking formulation. They construct and update a tree structure based on target trajectory. Then, when a query for a specific target is issued by the sink, this query is disseminated through the tree to search the node closest to the target.

The DOT, PET, and POOT algorithms are proposed to aid guided formulation. These algorithms mark nodes that detected the target, forming a path that will be followed by the tracker. In the algorithm proposed by Sharma et al. [2011b], the sink transmits the updated position of the target to the tracker directly by using a separate high power radio to enable the tracker to estimate the direction required for interception.

With the exception of the studies cited above, the algorithm classification obeys the push-based formulation of the tracking problem, in which a node detects a target and notifies the sink node.

6.3 Classification Based on Number of Targets

Several algorithms in sensor networks provided single target tracking [Bhuiyan et al. 2010; Zhang and Cao 2004a]. However, multi-target tracking (MTT) approaches are important due their real applications. Multi-target tracking consumes more energy because it increases the number of nodes employed in the tracking task, which also increases the number of samplings and reporting data packets [Hajiaghajani et al. 2012; Xu et al. 2004b].

The HCMTT [Hajiaghajani et al. 2012] minimizes redundant messages caused by targets that are close to each other by performing a merging mechanism. Thus, sensor nodes transmit data to a single node that generates a single report. In the DMMT [Hamouda and Phillips 2011], an algorithm selects the preferred target based on the influence strength of a target. This selection is performed when a sensor node detects more than one target. This sensor can track several targets simultaneously. Thus, a node locally decides which target it will track, leaving other targets for other sensors.

In the PES [Xu et al. 2004b] algorithm, it is assumed that nodes can identify all targets using a target code table. Thus, each target can be individually tracked. The OCO [Tran and Yang 2006] algorithm assumes the existence of two types of sensor nodes: the first distinguishes between targets, so it can accurately track each of the targets; the second detects whether or not there is a target within its sensing range (to activate its neighbors periodically).

The DAB, DAT, and DAT+Shortcut algorithms employ hierarchical structures to track a large number of targets. Nodes record information about the presence of targets. As a result, data about targets travel indiscriminately to the querying point, and the application can track more than one target simultaneously. In the evaluations of CODA, two continuous targets are tracked. In this case, tracking multiple continuous targets is simpler, because when they come close to each other, they behave as a single target.

The PET [Bhuiyan et al. 2010] and DCTC [Zhang and Cao 2004a] algorithms explicitly assume that only one target is being evaluated during the tracking task. The remaining algorithms do not evaluate and do not make it clear if they support multiple targets.

6.4 Classification Based on Type of Targets

There are two types of targets in the target-tracking literature: simple and continuous. Most studies aim to track simple targets, e.g., animals, persons, and vehicles. However, in some cases, it is desirable to track phenomena, such as forest fire, toxic

gases, biochemical material diffusion, and oil spill. These phenomena are called continuous targets, because they are continuously distributed across a region and usually cover a large field. In addition, they can diffuse, increase their size, change their shape, or split into multiple smaller targets [Chang et al. 2008; Hong et al. 2010; Park et al. 2010].

The CODA algorithm uses hybrid clustering to detect and track continuous target. For this purpose, the network is divided into static clusters. A continuous target may spread itself in more than one cluster. Then, the nodes that detect the continuous target notify their cluster heads. Then, the cluster heads estimate a set of boundary sensors that detect the target in each static cluster. Finally, these boundary sensors are united to form a dynamic group that represents the complete boundary of the target.

The PRECO [Hong et al. 2010] algorithm aims at prolonging sensor network lifetime, while tracking a continuous target. CODA [Chang et al. 2008] considers that all sensor nodes are activated even when most of the sensors are far from the target. For this reason, to optimize energy efficiency, node scheduling is used to activate just the nodes close to the continuous target. The TG-COD [Park et al. 2010] algorithm considers a two-tier grid structure: reactive fine-grained grid structures over a proactive coarse-grained grid structure. This approach seeks to reduce cluster organization cost and to provide the detailed shape of continuous targets.

7. FINAL REMARKS

This work presents a background study to answer some questions about target-tracking algorithms in WSNs such as: (1) What is target tracking? (2) What are the goals of the existing approaches for the tracking problem? (3) What the sensor network needs to perform target tracking efficiently? Briefly, the answers are:

- (1) Target tracking is the action of collecting sensor data about one or more targets, and estimating their positions in the sensor field.
- (2) The proposed target-tracking approaches aim at maintaining an accurate tracking and to report the results quickly while reducing communication costs and energy consumption.
- (3) To reach a satisfactory operation of the target-tracking application, some requirements have to be achieved. For instance, establishing a localization system, enabling the routing of data collected, performing density control, synchronizing sensor node clocks, and providing a secure communication service are key elements.

The main contribution of this work is the categorization of target-tracking solutions into three basic components that provide the basic mechanisms to detect the target and estimate its position, and three additional components that provide mechanisms to predict the position of target and use this information to save energy. Each of these components can be developed by using different techniques that affect the quality and efficiency of the tracking algorithm.

Another important contribution of this paper is the collection and classification of the state-of-the-art related to target tracking in sensor networks. Through this classification, we verified that in most of the existing tracking approaches, sensors

cooperate by using a cluster structure. The reason for this is that a cluster approach facilitates data fusion, and it can be dynamically adjusted in situations that require greater flexibility. On the other hand, tree structures are generally used when the tracking algorithm is based on queries. Finally, face structure is the most commonly applied technique in tracking approaches with guided vehicles.

The most appropriate structure depends on how the problem is formulated.

Regardless of the structure employed, the prediction algorithm associated with an awakening mechanism plays an important role on improving network lifetime, since prediction allows us to activate smaller sets of nodes to track the target.

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